Class 08: Breast Cancer Mini Proj

Ryan Ellis (PID: A17673864)

Table of contents

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

We will analyze a data set describing a fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can use the `read.csv()` function:

```
# IMPORTANT TO SET ROW NAMES EQUAL TO ONE
wisc.df<- read.csv("WisconsinCancer.csv", row.names=1)
```

```
head(wisc.df, 3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

We need to exclude the `diagnosis` column for analysis as this is the answer we are trying to achieve.

```
# This will give us everything but the first column
wisc.data <- wisc.df[,-1]
diagnosis<- as.factor(wisc.df$diagnosis)
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

There are 569 patients or observations.

Q2. How many of the observations have a malignant diagnosis?

```
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

There are 357 benign and 212 are malignant.

Q3. How many variables/features in the data are suffixed with _mean?

We can use the `grep()` function:

```
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"           
 [3] "perimeter_mean"          "area_mean"              
 [5] "smoothness_mean"         "compactness_mean"       
 [7] "concavity_mean"          "concave.points_mean"    
 [9] "symmetry_mean"           "fractal_dimension_mean" 
[11] "radius_se"               "texture_se"             
[13] "perimeter_se"            "area_se"                
[15] "smoothness_se"           "compactness_se"         
[17] "concavity_se"            "concave.points_se"      
[19] "symmetry_se"             "fractal_dimension_se"   
[21] "radius_worst"            "texture_worst"          
[23] "perimeter_worst"         "area_worst"             
[25] "smoothness_worst"        "compactness_worst"      
[27] "concavity_worst"         "concave.points_worst"   
[29] "symmetry_worst"          "fractal_dimension_worst"
```

```r
length(grep("_mean",colnames(wisc.data)))
```

```
[1] 10
```

There are 10 variables with `_mean`

Principle Component Analysis (PCA)

Need to scale data before PCA. `scale=TRUE` in the argument of `prcomp()`.

```r
## In general will always want to include scaling, will make the plots easier to interpret a

wisc.pr<- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

Around 44.27% of the data is captured via PC1

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

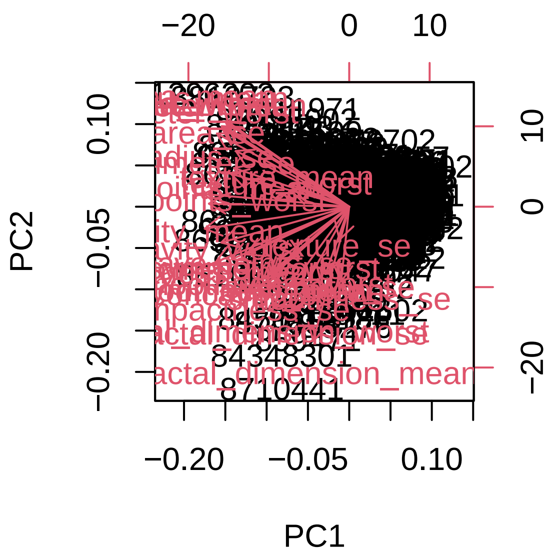
To describe 70% of the data you need at least the first three PC's so PC1-PC3

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

To get to 90% cumulative proportion it is necessary to have 7 principal components (PC1-PC7)

Lets visualize

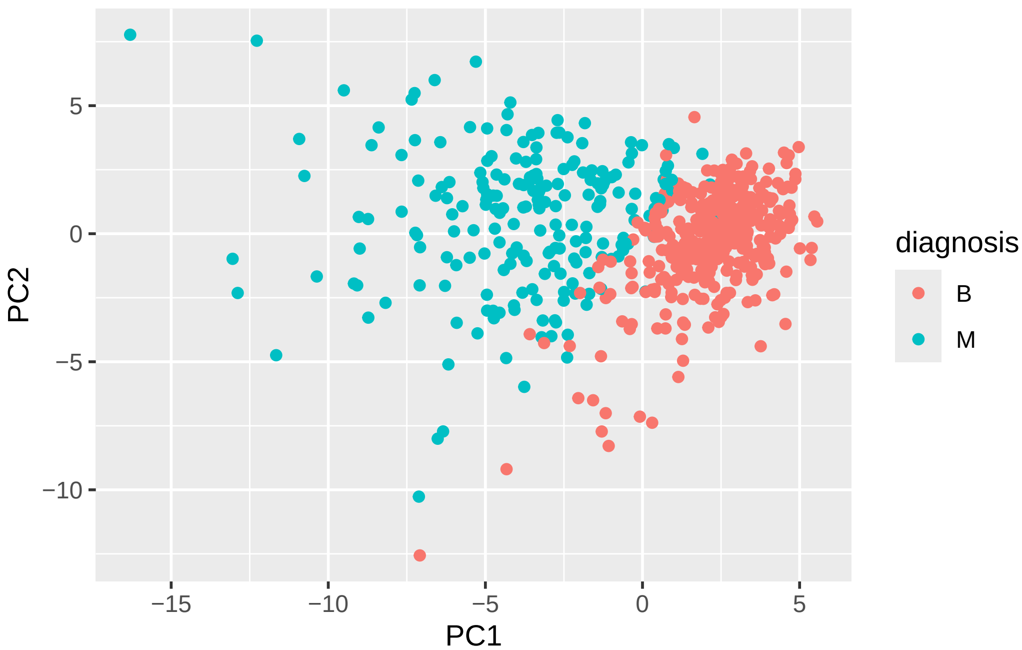
```
biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The sheer amount of data that this plot is trying to visualize. It is extremely difficult to visualize much of anything within the plot due to the number of points and different thing occurring.

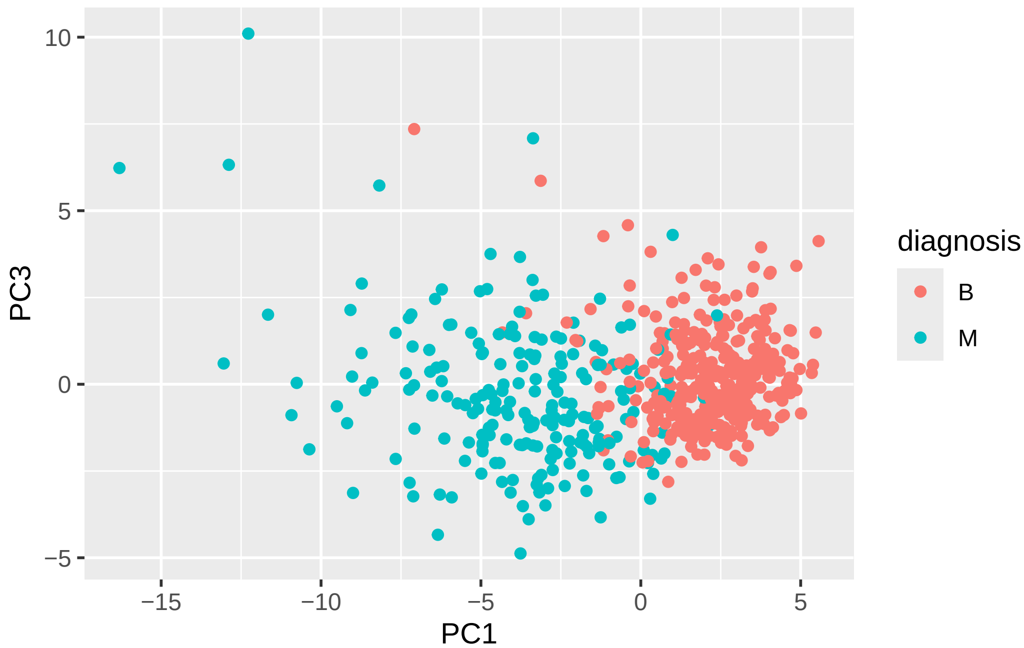
```
library(ggplot2)
ggplot(wisc.pr$x)+
  aes(PC1, PC2, col=diagnosis)+
  geom_point()
```



> Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

That both PC1/PC2 and PC1/PC3 are separated via diagnosis on PC1 axis, but both heavily overlap in PC2 and PC3. PC1 captures the most amount of variance.

```
ggplot(wisc.pr$x)+
  aes(PC1, PC3, col=diagnosis)+
  geom_point()
```



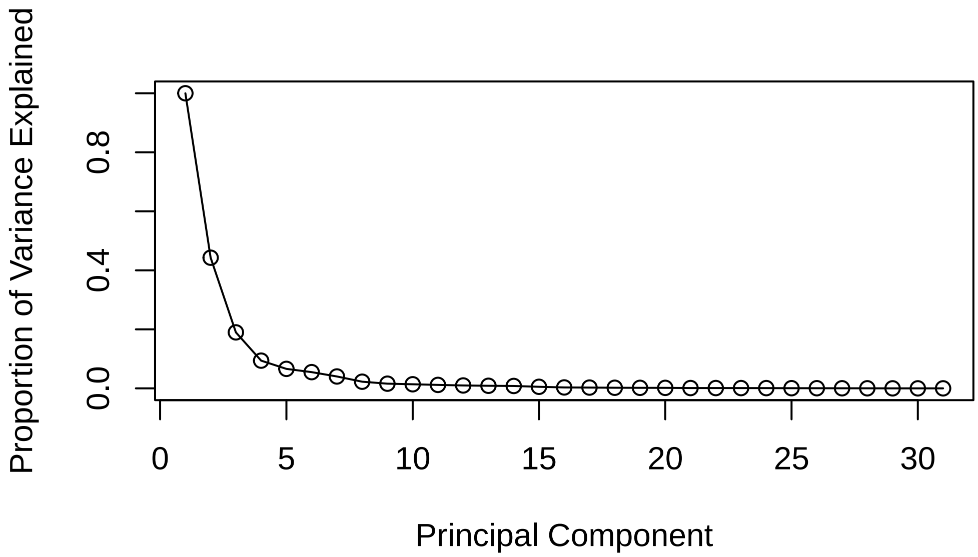
Varaince

We will make a scree plot, which will show us the variance that each PC indicates. Basically creates a exponential graph (hill), that will show when adding new PC's doesn't really make sense anymore.

```
pr.var<-(wisc.pr$sdev^2)
```

```
pve<- ((pr.var)/(30))

plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Commuicating PCA Results

Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

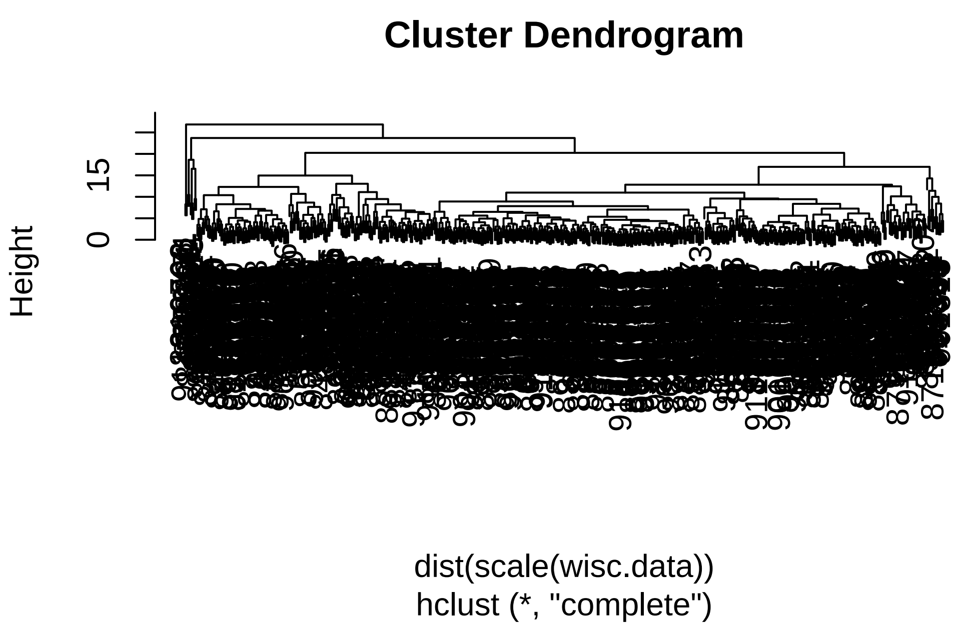
```
            radius_mean            texture_mean          perimeter_mean
            -0.21890244             -0.10372458             -0.22753729
              area_mean         smoothness_mean        compactness_mean
            -0.22099499             -0.14258969             -0.23928535
         concavity_mean     concave.points_mean           symmetry_mean
            -0.25840048             -0.26085376             -0.13816696
 fractal_dimension_mean               radius_se              texture_se
            -0.06436335             -0.20597878             -0.01742803
           perimeter_se                 area_se           smoothness_se
            -0.21132592             -0.20286964             -0.01453145
         compactness_se            concavity_se       concave.points_se
            -0.17039345             -0.15358979             -0.18341740
```

```
         symmetry_se    fractal_dimension_se            radius_worst 
         -0.04249842             -0.10256832             -0.22799663 
       texture_worst         perimeter_worst              area_worst 
         -0.10446933             -0.23663968             -0.22487053 
    smoothness_worst       compactness_worst         concavity_worst 
         -0.12795256             -0.21009588             -0.22876753 
concave.points_worst          symmetry_worst fractal_dimension_worst 
         -0.25088597             -0.12290456             -0.13178394 
```

The proportional component of PC1 that is `concave.points_mean` is -.261, and there is no other component with as much magnitude. This means that this feature has the biggest influence on where the point (patient) is placed in the PC1 axis.

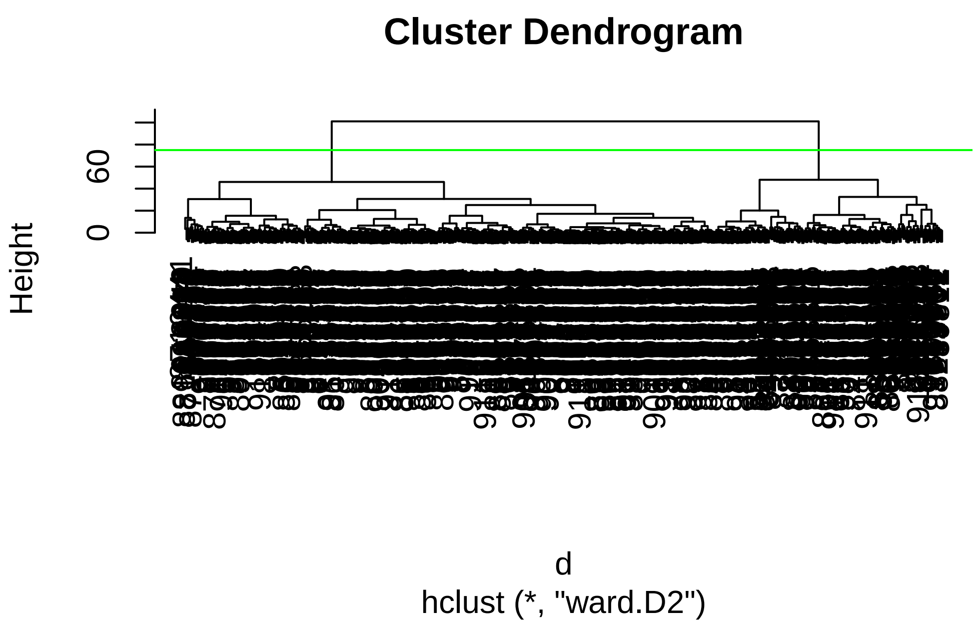
Hierarchical Clustering

```r
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```



Combine Methods

```r
d<-dist(wisc.pr$x[,1:4])
wisc.pr.hclust<- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=75, col="green")
```




```r
grps<- cutree(wisc.pr.hclust, h=75)
table(grps)
```

```
grps
  1   2
171 398
```

```r
# These results are similar to the diagnosis, where we cluster the two groups into Malignant
```

How does this clustering in `grps()` correspond to the expert `diagnosis`

```r
table(diagnosis)
```

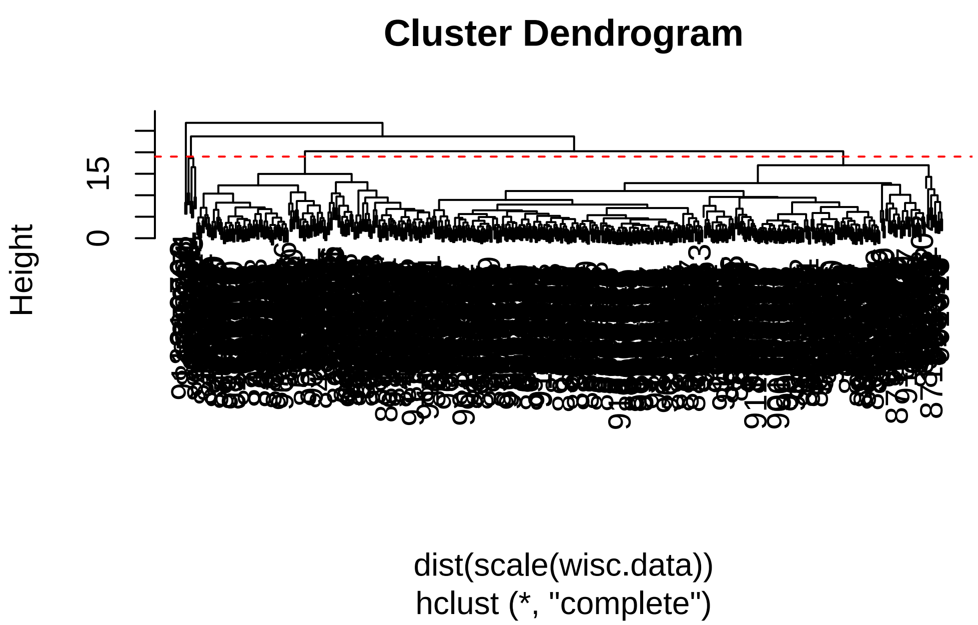
```
diagnosis
  B   M
357 212
```

```r
table(diagnosis, grps)
```

```
         grps
diagnosis   1   2
        B   6 351
        M 165  47
```

> Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

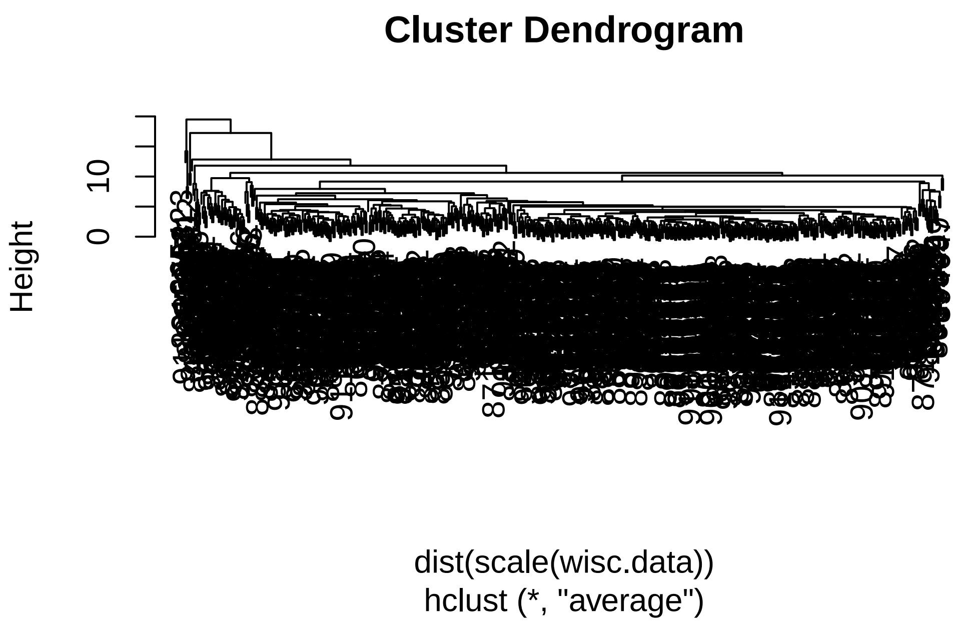
```r
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```



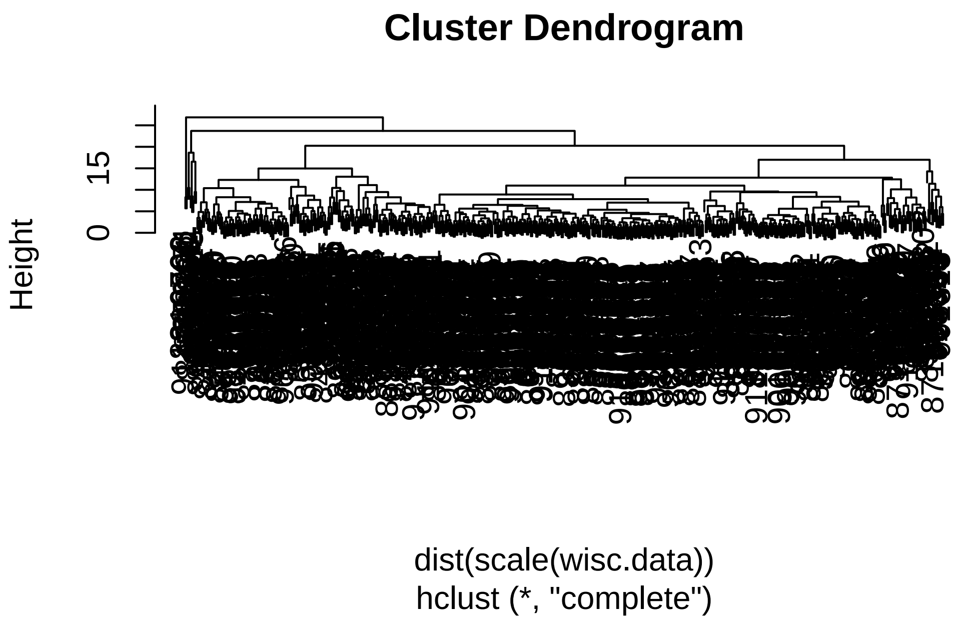
The hieght at which the normal clustering model has 4 clusters without PCA is around 19.

> Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

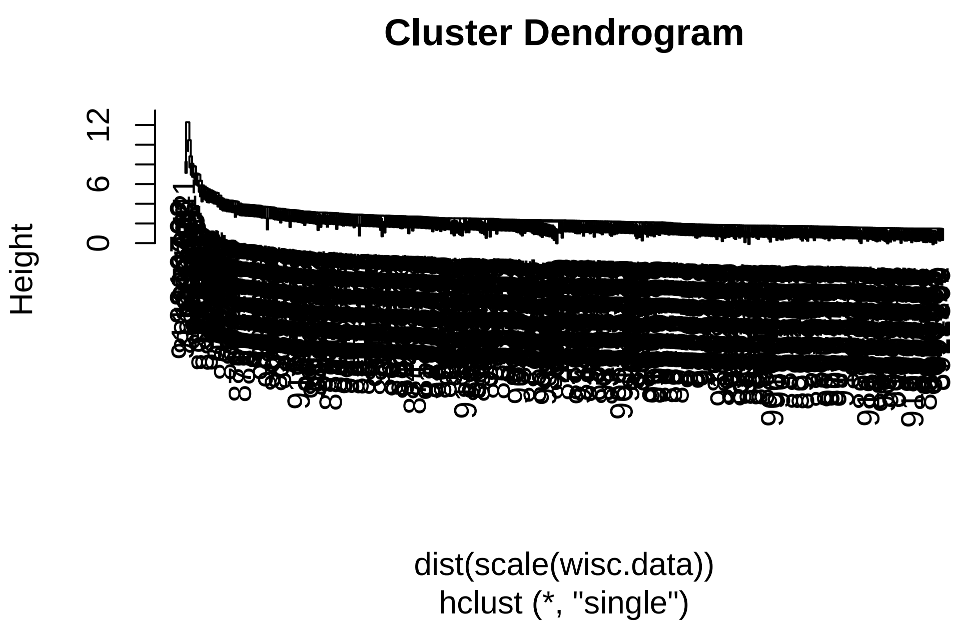
```r
method.hclust<- hclust(dist(scale(wisc.data)), method="average")
plot(method.hclust)
```



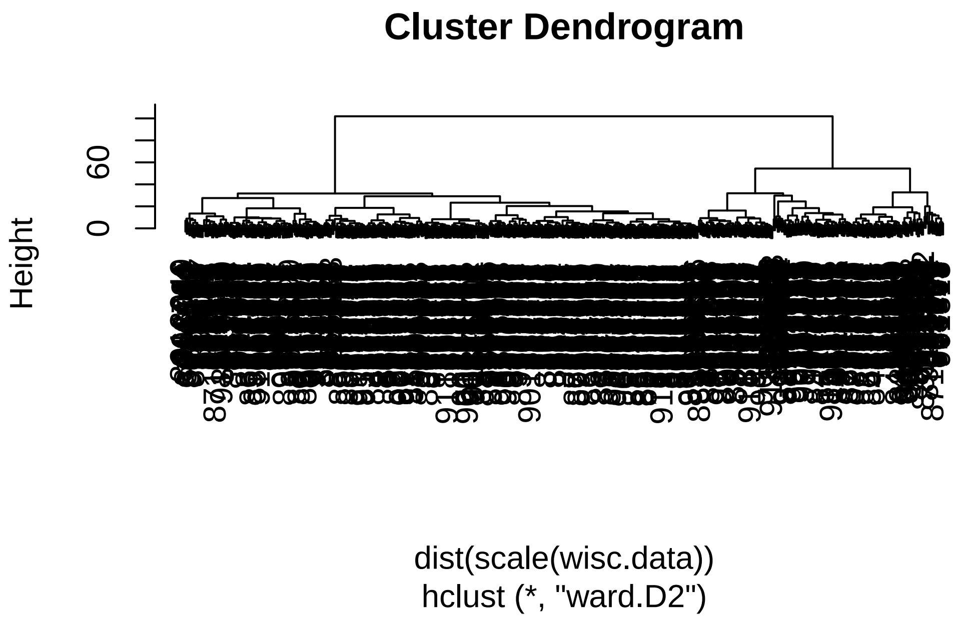

```r
method.hclust<- hclust(dist(scale(wisc.data)), method="complete")
plot(method.hclust)
```




```r
method.hclust<- hclust(dist(scale(wisc.data)), method="single")
plot(method.hclust)
```



```r
method.hclust<- hclust(dist(scale(wisc.data)), method="ward.D2")
plot(method.hclust)
```



The best method is `ward.D2`, also average is alright. They both aim to spread the data put so it is easier to interpret the clusters. but `ward.D2` does a much better job of this when looking at the above plots of the different methods.

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

It does decently when we compare it to the expert diagnosis, it shows that we have 6 false positive for benign and 47 false negatives for malignant. This is shown in the table above plotted grps versus diagnosis.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```r
wisc.hclust.clusters <- cutree(wisc.hclust, k=2)
table(diagnosis,wisc.hclust.clusters)
```

```
         wisc.hclust.clusters
diagnosis   1   2
        B 357   0
        M 210   2
```

Before using PCA, we can see that there is almost no separation between the two types, benign and malignant. Although when using PCA we have some false negative and positives, without it we cant really even interpret anything.

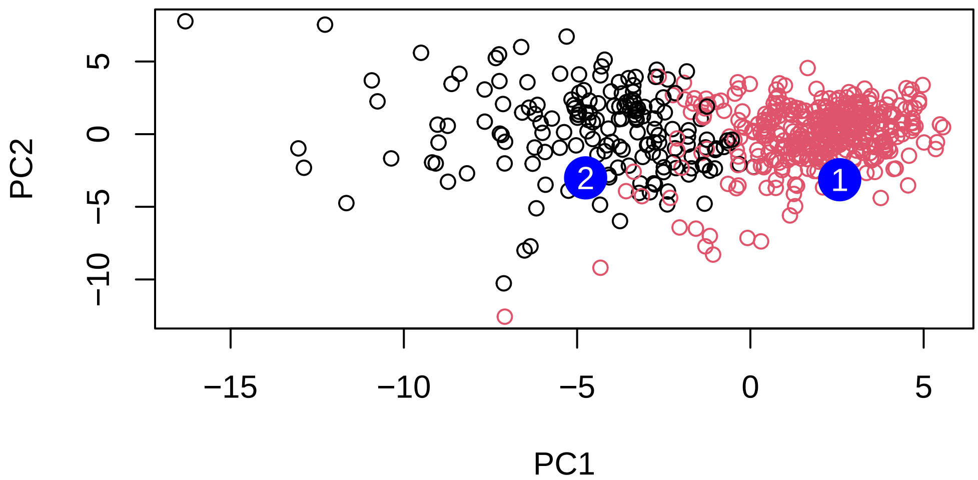
Prediction

We will now utilize the `predict()` function to combine our previous PCA model and and some new data (cancer cell data)

```r
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

When viewing the plot, and based off the previous conclusions drawn from the PCA plots, a malignant patient(tumor), is likely to be in the negative portion on the x axis (PC1). Thus when comparing patient 1 and 2 because patient 2 lies on the negative side amoungst the malignant patients I would predict this patient to be prioritized.